Lesson 10

ENVELOPE THERMAL PERFORMANCE

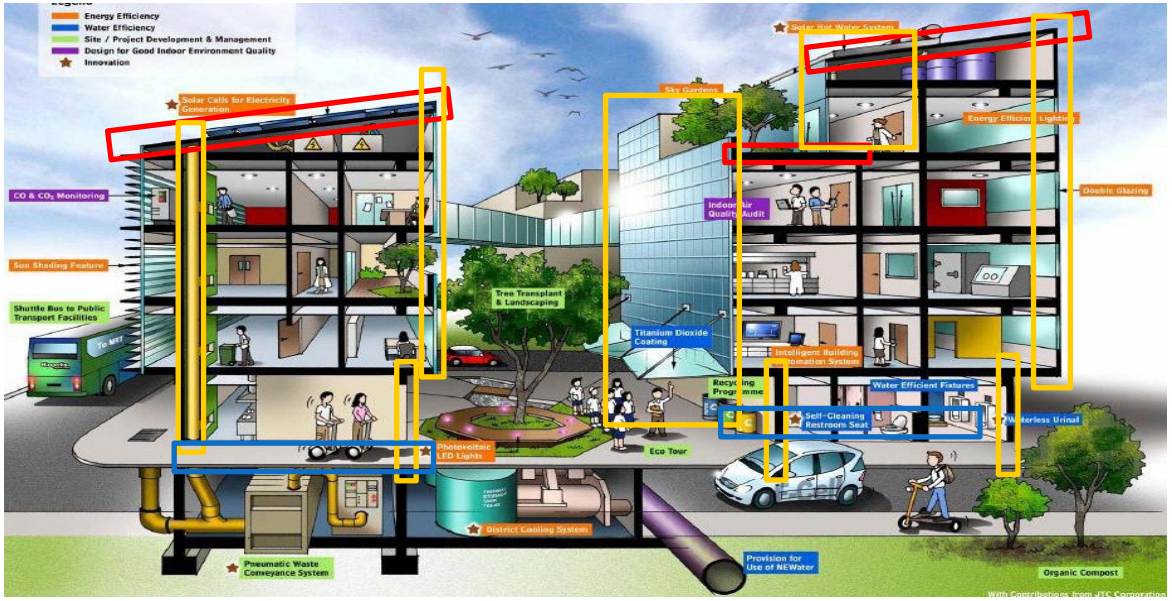
Learning Outcomes

* Understand how design of the building envelope affects heat gains into buildings
* Explain some passive cooling strategies and how to apply them to reduce building heat gains.
* Understand ETTV, RTTV, RETV and Roof insulation concepts and perform ETTV calculations

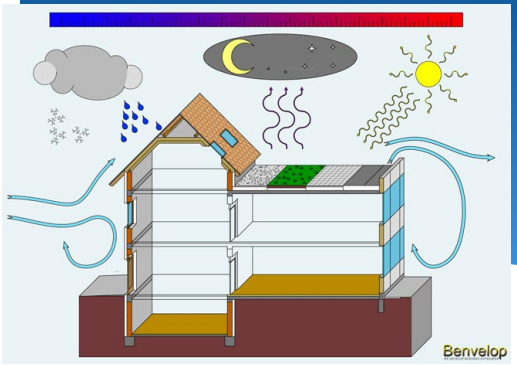
Building Envelope

* Separation between interior and exterior environments
* Roof, external walls including windows and floors.

Roof External Walls Windows, Doors



Floor

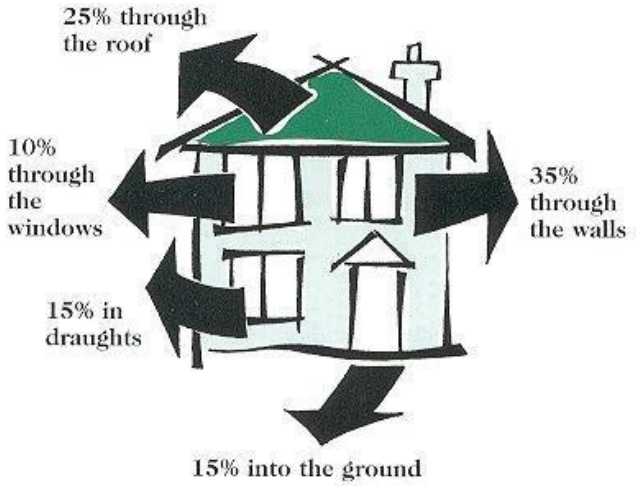


Building Envelope

- Protect and facilitates control of indoor climate.
- Keeps out moisture and insulates against heat losses and gains, while admitting useful daylight.
- Influences building performance - thermal comfort, daylighting, glare, energy efficiency, natural ventilation and noise.
- Performance analysed by reaction of its components to heat, moisture and air transfer.

Code on Envelope Thermal Performance for Buildings

Code on Envelope Thermal Performance for Buildings



Sufficiently Insulated Envelopes (A/C Buildings)

- Envelope Thermal Transfer Value (ETTV)
- Roof Thermal Transfer Value (RTTV)
- Residential Envelope Transmittance Value (RETV)
- Roof Insulation

ETTV

GM 2021

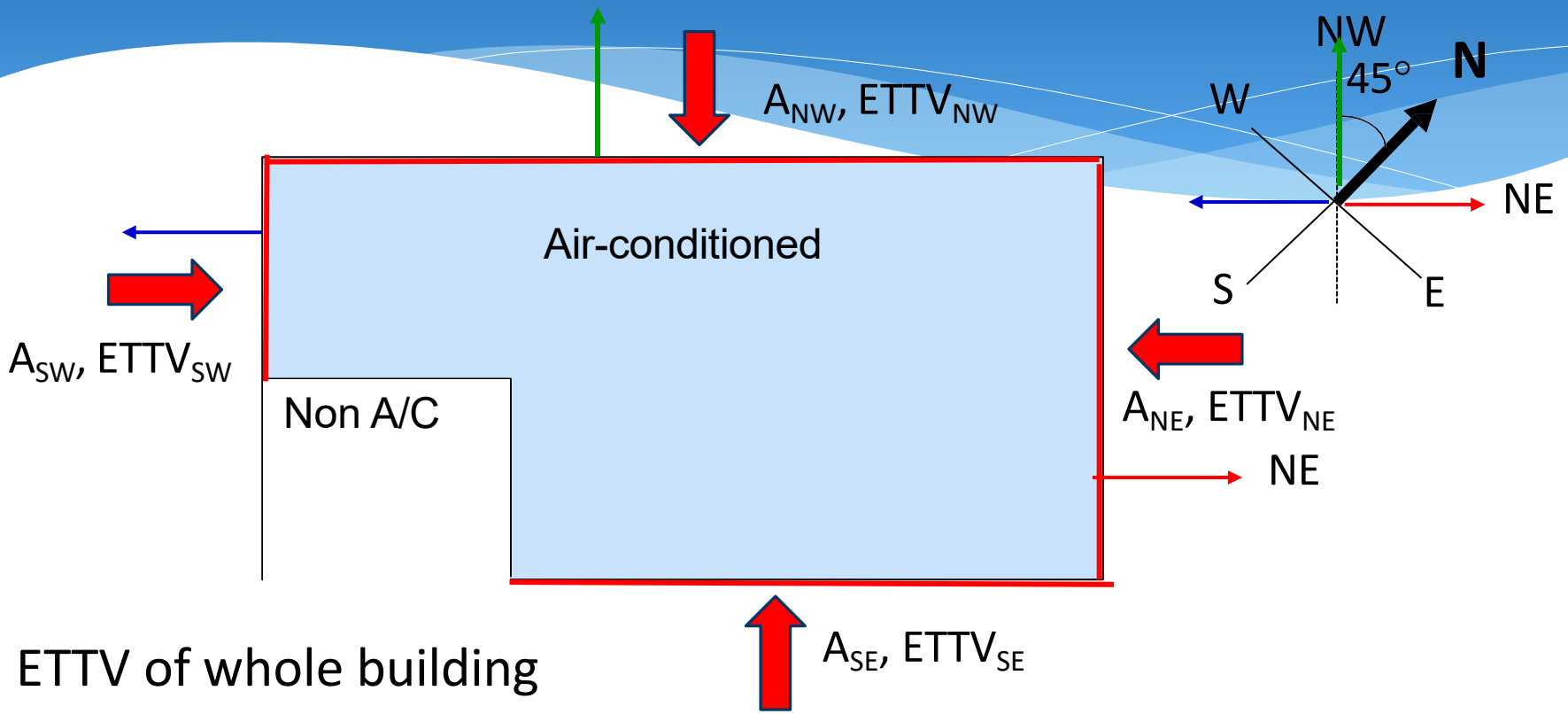
A measure of average heat gain through external walls enclosing air-conditioned spaces in a non-residential building.

- Heat conduction through opaque walls
- Heat conduction through glass windows
- Solar radiation through glass windows



Max permissible ETTV - **50 W/m²**

ETTV of Façade & Whole Building



ETTV of whole building

$$= \frac{A_{NE}*ETTV_{NE} + A_{SE}*ETTV_{SE} + A_{SW}*ETTV_{SW} + A_{NW}*ETTV_{NW}}{A_o}$$

Max permissible ETTV - **50 W/m²**

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ETTV for a Facade

Opaque wall - conduction

Glass window - conduction

Glass window – solar radiation

$$ETTV = 12\ (1\text{-}WWR)U_w + 3.4(WWR)\ U_f + 211(WWR)(CF)(SC)$$

Where:

ETTV = envelope thermal transfer value (W/m^2)

WWR = window-to-wall ratio (fenestration area/gross area of exterior wall)

U_w = thermal transmittance of opaque walls (W/m^2K)

U_f = thermal transmittance of fenestration (W/m^2K)

CF = correction factor for solar heat gain through fenestration

SC = shading coefficients of fenestration

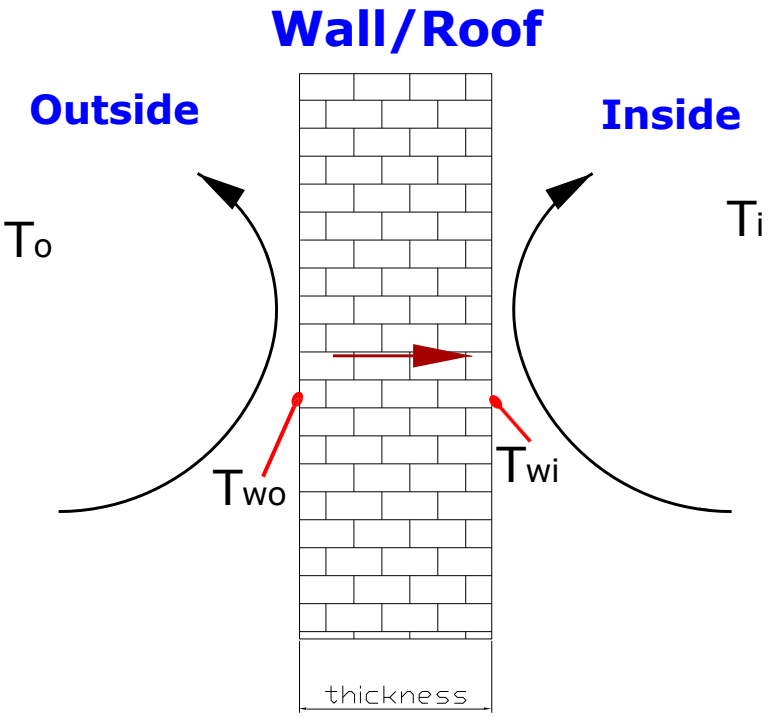
Thermal Transmittance (U-Value)

Rate of heat conducted through the wall/roof can be calculated using **air temperatures**:

$Q = U_T * A * (T_o - T_i)$ Watts

Where:

U_T = thermal transmittance, W/m^2K
A = wall/roof surface area, m^2
T_o = outside air temperature, °C
T_i = inside air temperature, °C

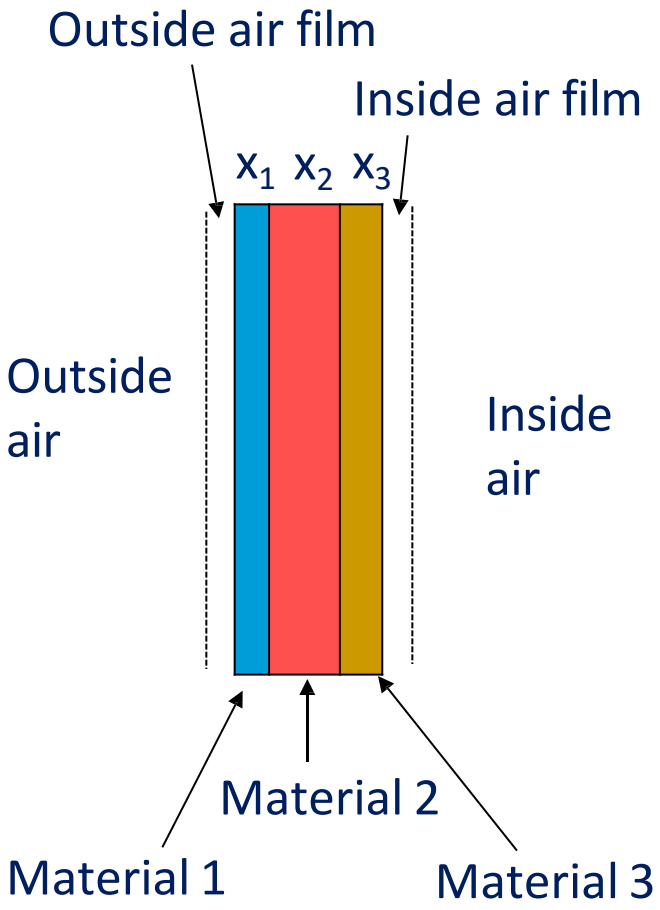


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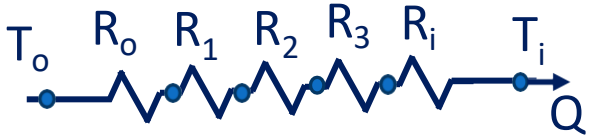
Thermal Transmittance (U-Value)

$$R_T = \frac{1}{h_o} + \frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{x_3}{k_3} + \frac{1}{h_i}$$

$$= R_o + R_1 + R_2 + R_3 + R_i \quad m^2K/W$$



$$U_T = \frac{1}{R_T}$$



Where

h_o = outside surface convective heat transfer coefficient, W/m^2K

h_i = inside surface convective heat transfer coefficient, W/m^2K

R_o = outside surface air film resistance, m^2K/W

R_i = inside surface air film resistance, m^2K/W

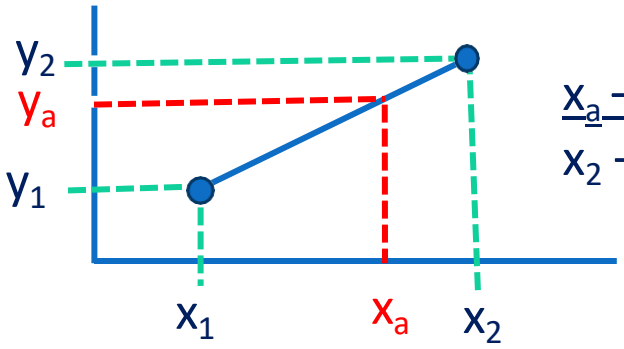
Thermal Transmittance (U-Value)

If an air gap is used in between layers of materials, then total thermal resistance is

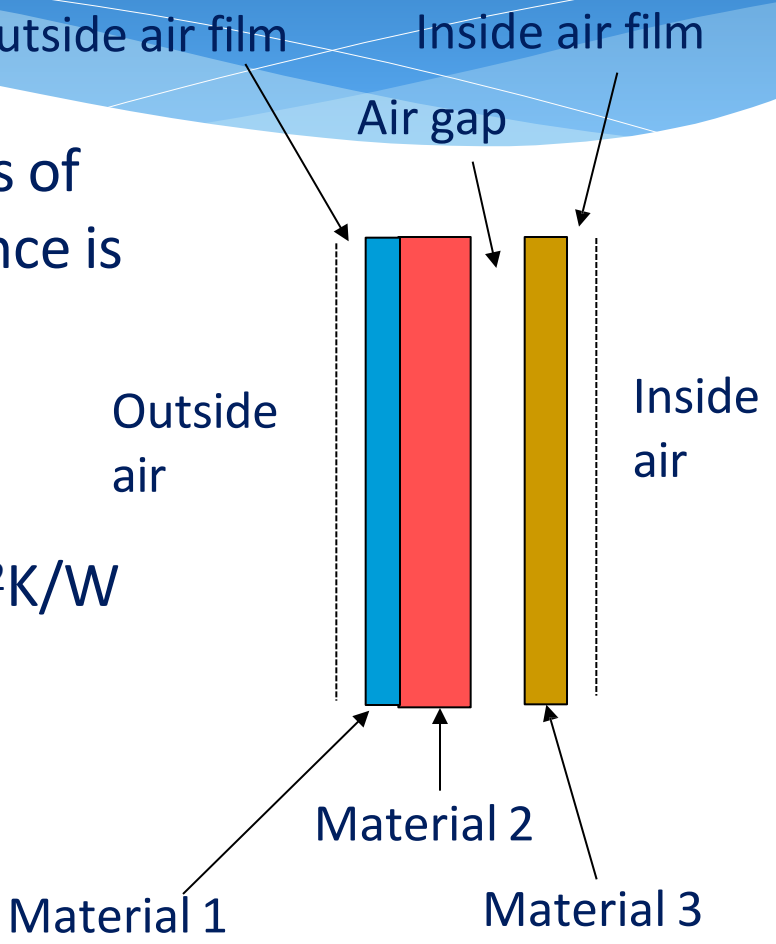
$R_T = R_o + R_1 + R_2 + R_a + R_3 + R_i$

Where

R_a = thermal resistance of air gap, m^2K/W



$$\frac{x_a - x_1}{x_2 - x_1} = \frac{y_a - y_1}{y_2 - y_1}$$



K-Value

Material	Density (kg/m^3)	K-value (W/mK)
Concrete	2400 64	1.442 0.144
Plaster: (a) gypsum (b) perlite (c) sand/cement (d) vermiculite	 1216 616 1568 640 - 960	 0.370 0.115 0.533 0.202 – 0.303

Table C5: Thermal Conductivity Values (k-Values) of Basic Materials

S/No	Material	Density (kg/m^3)	k-value (W/m K)
1	Asphalt, roofing	2240	1.226
2	Bitumen	-	1.298
3	Brick		
	(a) dry (covered by plaster or tiles outside)	1760	0.807
	(b) common brickwall (brickwall directly exposed to weather outside).	-	1.154
4	Concrete	2400	1.442
		64	0.144
5	Concrete, lightweight	960	0.303
		1120	0.346
		1280	0.476
6	Cork board	144	0.042
7	Fibre board	264	0.052

8	Fibre glass (See Glass Wool and Mineral Wool)		
9	Glass, sheet	2512	1.053
10	Glass wool, mat or guilt (dry)	32	0.035
11	Gypsum plaster board	880	0.170
12	Hard board:		
	(a) Standard	1024	0.216
	(b) medium	640	0.123
13	Metals:		
	(a) Aluminium alloy, typical	2672	211
	(b) Copper, commercial	8784	385
	(c) Steel	7840	47.6
14	Mineral wool, felt	32 - 104	0.035 - 0.032
15	Plaster:		
	(a) gypsum	1216	0.370
	(b) perlite	616	0.115
	(c) sand / cement	1568	0.533
	(d) vermiculite	640 - 960	0.202 - 0.303

16	Polystyrene, expanded	16	0.035
17	Polyurethane, foam	24	0.024
18	PVC flooring	1360	0.713
19	Soil, loosely packed	1200	0.375
20	Stone, tile:		
	(a) sand stone	2000	1.298
	(b) granite	2640	2.927
	(c) marble / terrazzo / ceramic / mosaic	2640	1.298
21	Tile, roof	1890	0.836
22	Timber:		
	(a) across grain softwood	608	0.125
	(b) hardwood	702	0.138
	(c) plywood	528	0.138
23	Vermiculite, loose granules	80-112	0.065
24	Wood chipboard	800	0.144
25	Woodwool slab	400	0.086
		480	0.101

Surface Film Resistances

Type of Surface	Thermal Resistance (m^2 K/W)
Surface Film Resistances for Walls	
Inside surface (R_i)	
(a) High emissivity	0.120
(b) Low emissivity	0.299
Outside surface (R_o)	
- High emissivity	0.044

Example

Calculate the thermal transmittance, U, value of a wall constructed of 150 mm concrete (2400 kg/m^3) and one layer of 12 mm thick sand/cement plaster on the outside and inside surfaces. Assume the inside surface of the opaque wall has low emissivity.

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Example - Solution

$$R_T = R_o + \frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{x_3}{k_3} + R_i$$

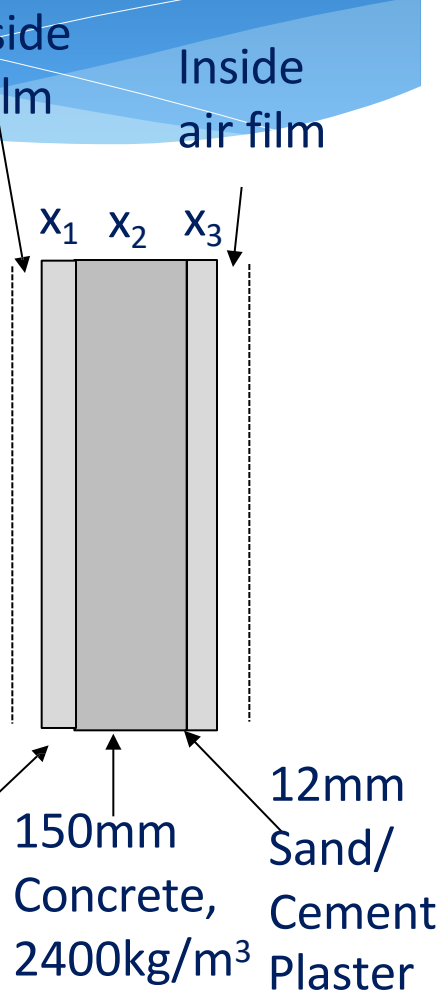
From Fig 3.2, $R_o = 0.044\ m^2K/W$ & $R_i = 0.299\ m^2K/W$

From Fig 3.1, $k_1 = k_3 = 0.533 W/mK$, $k_2 = 1.442 W/mK$

$$R_T = 0.044 + \frac{0.012}{0.533} + \frac{0.15}{1.442} + \frac{0.012}{0.533} + 0.299$$

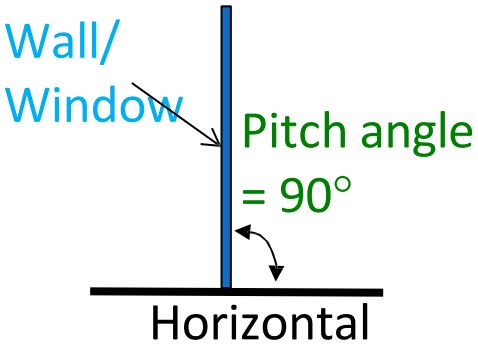
$$= 0.492\ m^2K/W$$

$$U_T = \frac{1}{R_T} = \frac{1}{0.492} = 2.033\ W/m^2K$$



Solar Correction Factor – Walls (ETTV)

Solar Correction factor (CF) for fenestration at different orientations, e.g. N, E, S, W, is amount of solar radiation it receives compare with average value received by vertical walls at eight orientations.



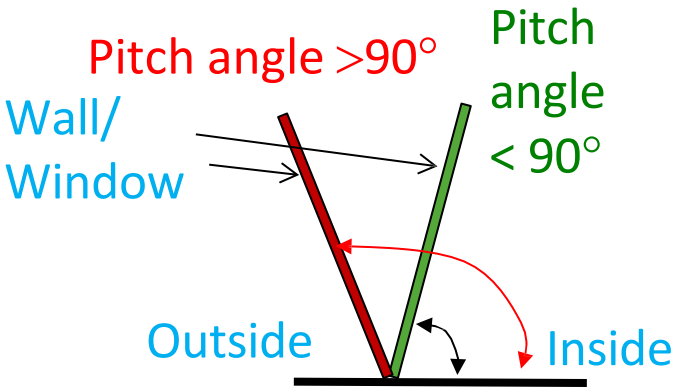


Table C1: Solar Correction Factors (CF) for Walls (ETTV)

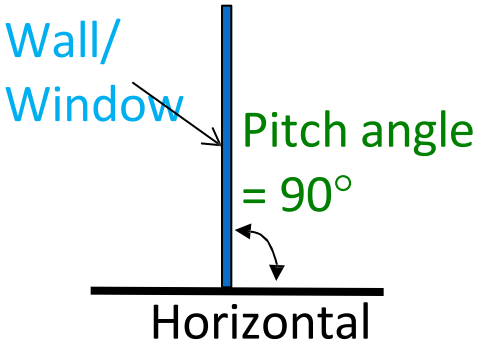
Pitch Angle	Orientation							
	N	NE	E	SE	S	SW	W	NW
70°	1.17	1.33	1.47	1.35	1.21	1.41	1.56	1.38
75°	1.07	1.23	1.37	1.25	1.11	1.32	1.47	1.28
80°	0.98	1.14	1.30	1.16	1.01	1.23	1.39	1.20
85°	0.89	1.05	1.21	1.07	0.92	1.14	1.31	1.11
90°	0.80	0.97	1.13	0.98	0.83	1.06	1.23	1.03
95°	0.73	0.90	1.05	0.91	0.76	0.99	1.15	0.96
100°	0.67	0.83	0.97	0.84	0.70	0.92	1.08	0.89
105°	0.62	0.77	0.90	0.78	0.65	0.86	1.01	0.83
110°	0.59	0.72	0.83	0.72	0.61	0.80	0.94	0.78
115°	0.57	0.67	0.77	0.67	0.58	0.75	0.87	0.73
120°	0.55	0.63	0.72	0.63	0.56	0.71	0.81	0.69

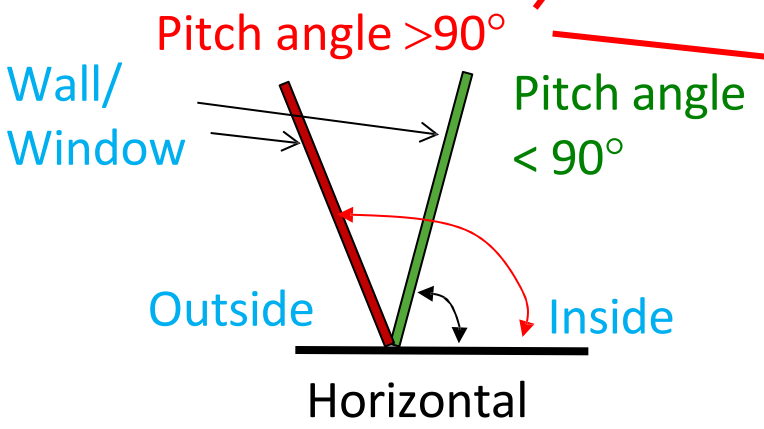
Note: 1. The correction factors for other orientations and other pitch angles may be obtained by interpolation.

Correction Factor



Pitch angle
< 90°







Shading Coefficients

Shading coefficient is derived from the annual average solar radiation transmitted through an unshaded 3mm clear window glass.

For other types of fenestration system, the shading coefficient (SC) is given by:

$$SC = \frac{Solar\ heat\ gain\ of\ any\ glass\ and\ shading\ combination}{Solar\ heat\ gain\ through\ a\ 3mm\ unshaded\ clear\ glass}$$

Shading coefficient for any fenestration system

= **SC of the glass** x **SC of sun shading devices**.

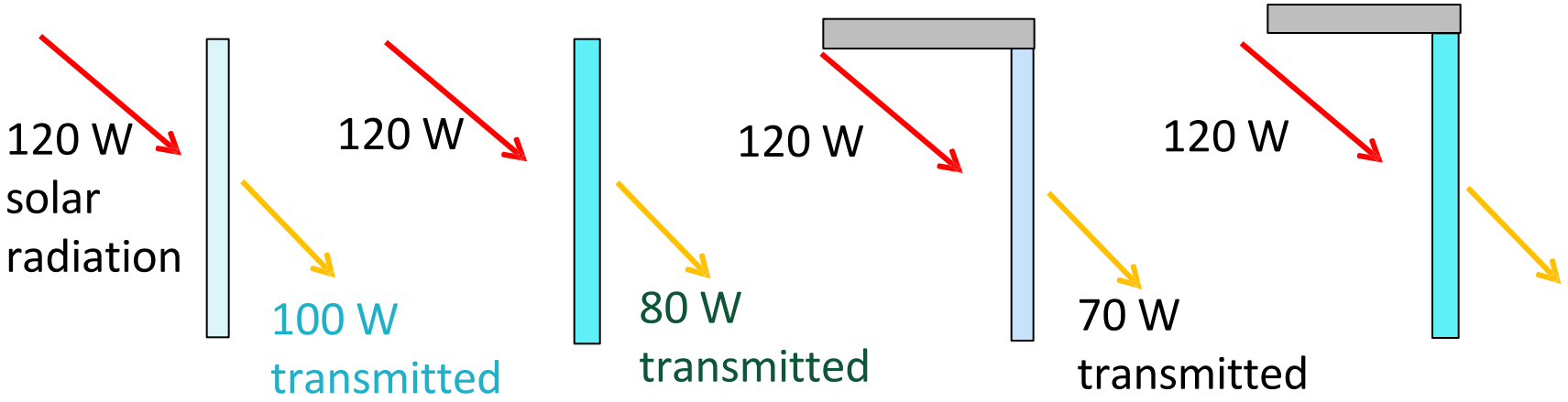
Shading Coefficient

3mm clear glass

10 mm tinted glass

3mm clear glass with external shading

10 mm tinted glass with same external shading



SC of 10 mm tinted glass,

$SC_1 = \frac{80}{100} = 0.8$

SC of external shading,

$SC_2 = \frac{70}{100} = 0.7$

SC of 10 mm tinted glass with same external shading = $SC_1 * SC_2 = 0.8 * 0.7 = 0.56$

Combination of Façade Materials & Fenestration

$$ETTV = \frac{12(A_{w1} \times U_{w1} + A_{w2} \times U_{w2} + \ldots + A_{wn} \times U_{wn})}{A_o} +$$

$$\frac{3.4(A_{f1} \times U_{f1} + A_{f2} \times U_{f2} + \ldots + A_{fn} \times U_{fn})}{A_o} +$$

$$\frac{211(A_{f1} \times SC_{f1} + A_{f2} \times SC_{f2} + \ldots + A_{fn} \times SC_{fn})(CF)}{A_o}$$

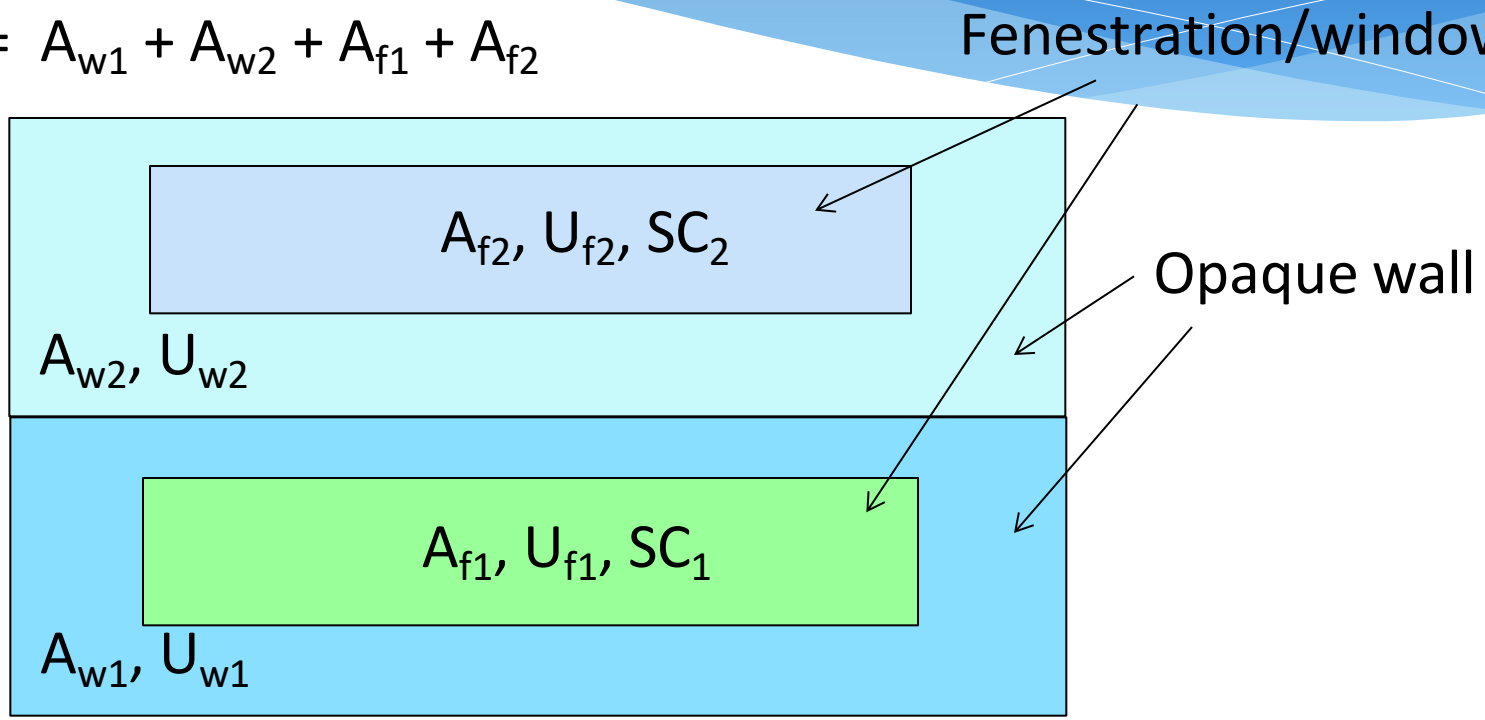
where

A_{w1}, A_{w2}, A_{wn}	:	areas of different opaque wall (m^2)
A_{f1}, A_{f2}, A_{fn}	:	areas of different fenestration (m^2)
A_o	:	gross area of the exterior wall (m^2)
U_{w1}, U_{w2}, U_{wn}	:	thermal transmittances of opaque walls ($W/m^2\,°K$)
U_{f1}, U_{f2}, U_{fn}	:	thermal transmittances of fenestrations ($W/m^2\,°K$)
SC_{f1}, SC_{f2}, SC_{fn}	:	shading coefficients of fenestrations

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Combination of Façade Materials and Fenestration

$A_o = A_{w1} + A_{w2} + A_{f1} + A_{f2}$



ETTV of façade

$$= \frac{12(A_{w1}U_{w1} + A_{w2}U_{w2}) + 3.4(A_{f1}U_{f1} + A_{f2}U_{f2}) + 211(A_{f1}SC_{f1} + A_{f2}SC_{f2})CF}{A_o}$$

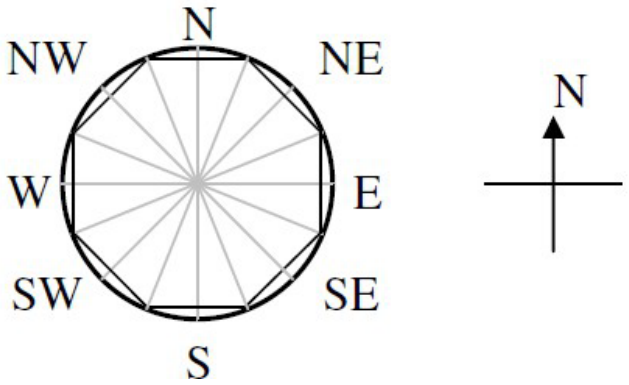
ETTV for Whole Building

$$ETTV = \frac{A_{o1} \times ETTV_1 + A_{o2} \times ETTV_2 + \ldots + A_{on} + ETTV_n}{A_{o1} + A_{o2} + \ldots + A_{on}}$$

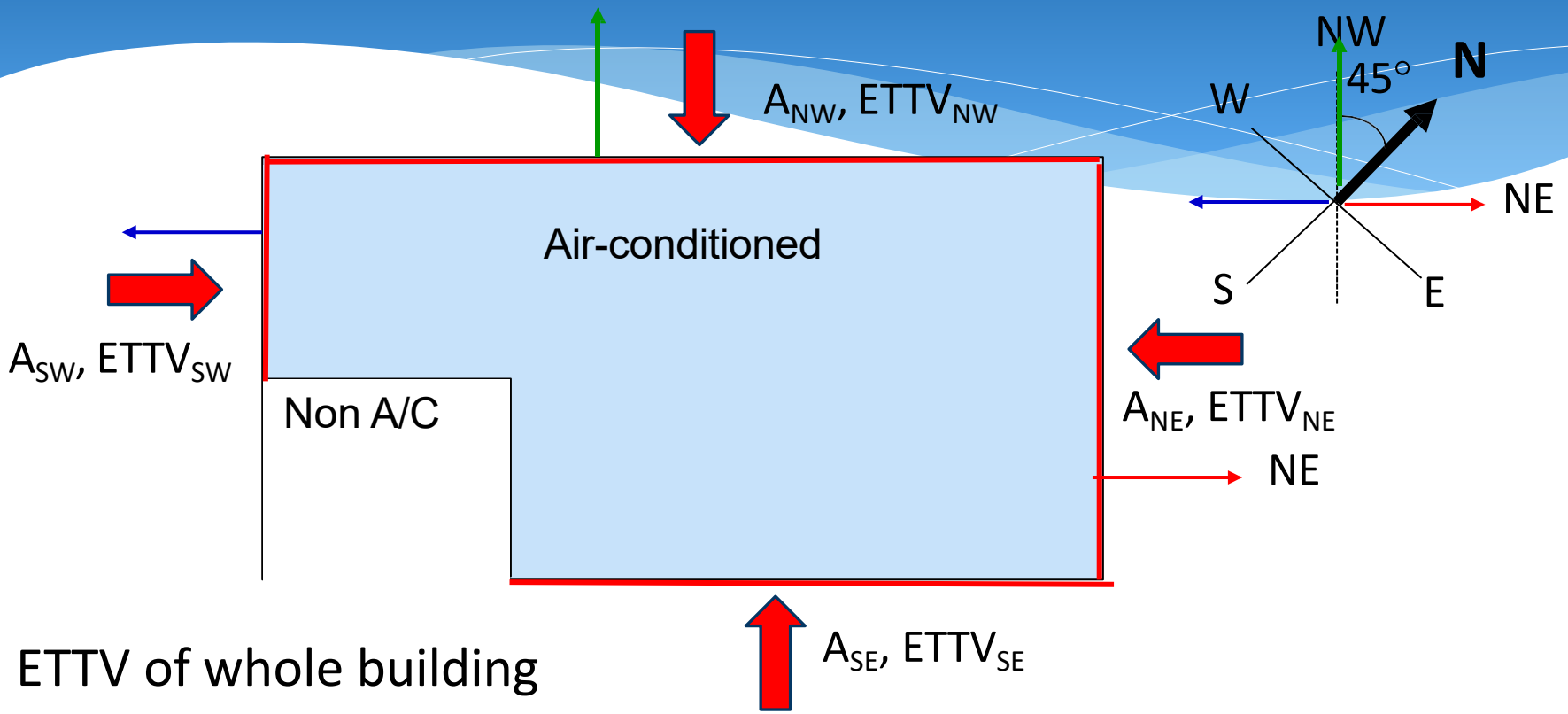
where

A_{o1}, A_{o2}, A_{on} : gross areas of the exterior wall for each orientation (m^2)

5.4.1 If the wall is curved, the eight primary orientations are segmented as follows:



ETTV of Façade & Whole Building



ETTV of whole building

$$= \frac{A_{NE}*ETTV_{NE} + A_{SE}*ETTV_{SE} + A_{SW}*ETTV_{SW} + A_{NW}*ETTV_{NW}}{A_o}$$

Max permissible ETTV - **50 W/m²**

ETTV of Whole Building

Facade	Façade Area	Façade ETTV	Façade Area x Façade ETTV
North East	$A_{NE} = 100$	$ETTV_{NE} = 40$	4000
South East	$A_{SE} = 150$	$ETTV_{SE} = 35$	5250
South West	$A_{SW} = 200$	$ETTV_{SW} = 55$	11000
North West	$A_{NW} = 100$	$ETTV_{NW} = 50$	5000
Total	**550**		**25250**
ETTV of Whole Building =		25250 ÷ 550 = 45.91 W/m^2	

Roof Insulation

Weight Group	Weight Range (kg/ m^2)	Maximum Thermal Transmittance (W/m^2K)
Light	Under 50	0.5
Medium	50 to 230	0.8
Heavy	Over 230	1.2

The average weight of a roof is calculated using the formula:

$$W_r = \frac{A_{r1} \times W_{r1} + A_{r2} \times W_{r2} + \ldots\ldots + A_{rn} \times W_{rn}}{A_{r1} + A_{r2} + \ldots\ldots + A_{rn}}$$

Where:

W_r = average weight of the roof (kg/m^2)

W_{r1}, … W_{rn} = respective weight of different roof sections (kg/m^2)

A_{r1}, … A_{rn} = respective area of the different roof sections (m^2)

More Than One Type of Roof

$$U_r = \frac{A_{r1} \times U_{r1} + A_{r2} \times U_{r2} + \ldots\ldots + A_{rn} \times U_{rn}}{A_{r1} + A_{r2} + \ldots\ldots + A_{rn}}$$

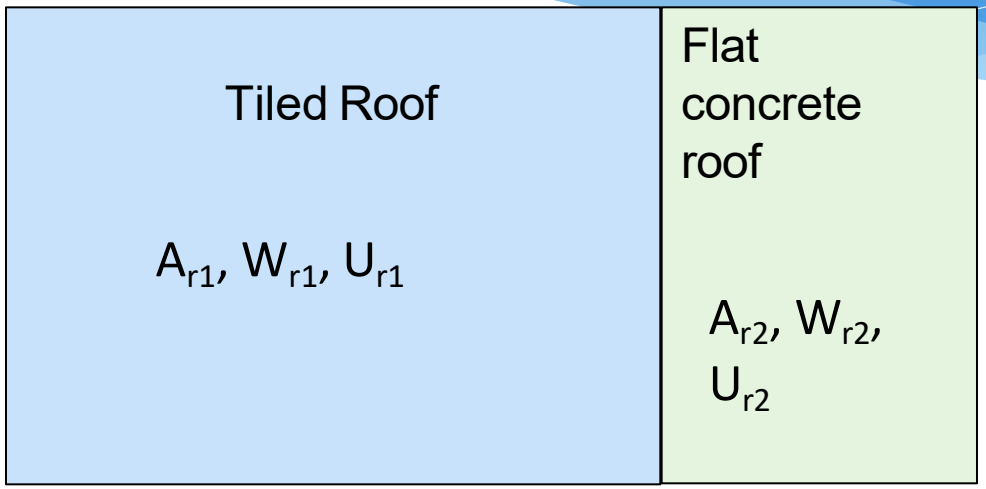
Where:

U_r = average thermal transmittance of the gross roof area (W/m^2K)

$U_{r1}, \ldots U_{rn}$ = respective thermal transmittance of different roof sections, (W/m^2K)

$A_{r1}, \ldots A_{rn}$ = respective area of different roof sections (m^2)

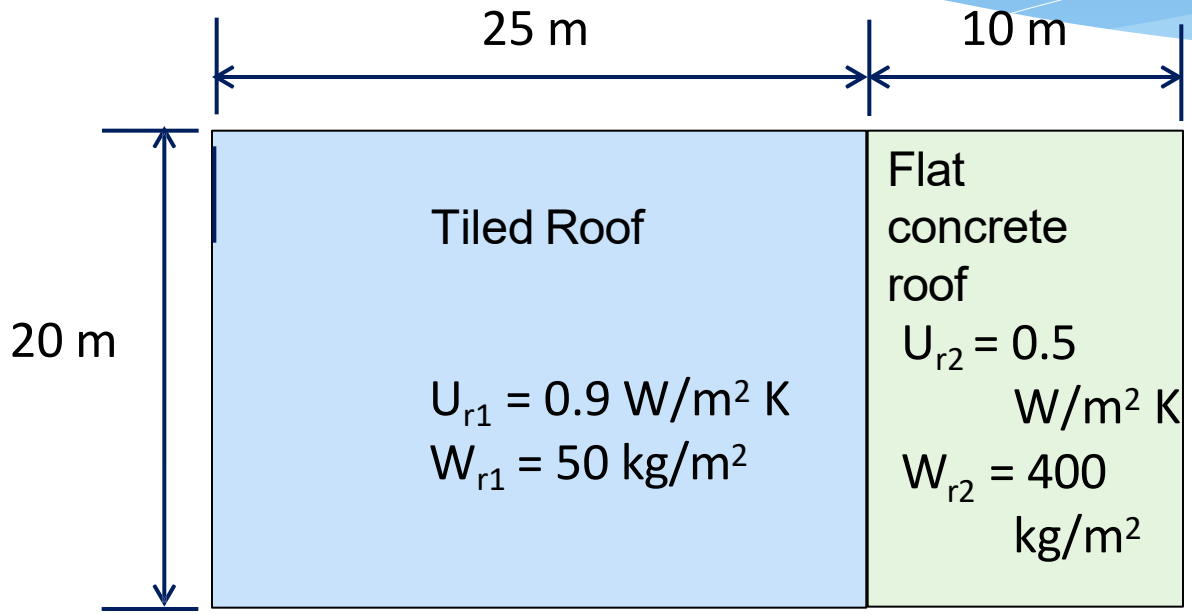
Roof Insulation



Average weight of 1 m^2 of roof, $W_r = \frac{A_{r1}*W_{r1} + A_{r2}*W_{r2}}{A_{r1} + A_{r2}}$

Average U-value of roof, $U_r = \frac{A_{r1}*U_{r1} + A_{r2}*U_{r2}}{A_{r1} + A_{r2}}$

Example



Does the roof comply with the U-value requirement?

Example - Solution

Average weight of 1 m^2 of roof, $W_r = \frac{A_{r1}*W_{r1} + A_{r2}*W_{r2}}{A_{r1} + A_{r2}}$

Average weight of 1 m^2 of roof, $W_r = \frac{(25x20)*50 + (10x20)*400}{(25x20) + (10x20)}$

$= 150\ kg/m^2$

Ave U-value of roof, $U_r = \frac{A_{r1}*U_{r1} + A_{r2}*U_{r2}}{A_{r1} + A_{r2}}$

Weight Group	Weight Range (kg/ m^2)	Maximum Thermal Transmittance (W/m^2K)
Light	Under 50	0.5.
Medium	50 to 230	0.8
Heavy	Over 230	1.2

Ave U-value of roof, $U_r = \frac{(25x20)*0.9 + (10x20)*0.5}{(25x20) + (10x20)}$

$= 0.786\ W/m^2K$

RTTV

A measure of average heat gain through the roof enclosing air-conditioned spaces in a non-residential building.





Heat conduction through opaque roof

Heat conduction through skylight

Solar radiation through skylight

Max permissible RTTV for air-conditioned non-residential buildings - **50 W/m²**

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RTTV

$$ETTV = 12\ (1\text{-}WWR)U_w + 3.4(WWR)\ U_f + 211(WWR)(CF)(SC)$$

$$RTTV = 12.5\ (1 - SKR)U_r + 4.8(SKR)\ U_s + 485(SKR)(CF)(SC)$$

Where:

RTTV = roof thermal transfer value (W/m^2)
SKR = skylight ratio of roof (skylight area/gross area of roof)
U_r = thermal transmittance of opaque roof (W/m^2K)
U_s = thermal transmittance of skylight area (W/m^2K)
CF = solar correction factor for roof
SC = shading coefficient of skylight portion of the roof

RTTV – Surface Film Resistances for Roofs

B. Surface Film Resistances for Roofs	
1. Inside surface (R_i)	
(a) High Emissivity	
(i) Flat roof	0.162
(ii) Sloped roof 22½°	0.148
(iii) Sloped roof 45°	0.133
(b) Low Emissivity	
(i) Flat roof	0.801
(ii) Sloped roof 22½°	0.595
(iii) Sloped roof 45°	0.391
2. Outside surface (R_o) - High emissivity Flat or sloped	0.055

Solar Correction Factor - Roofs

Table C2: Solar Correction Factors for Roofs

Pitch Angle	Orientation							
	N	NE	E	SE	S	SW	W	NW
0°	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
5°	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
10°	0.99	0.99	1.00	1.00	1.00	0.99	0.99	0.99
15°	0.98	0.98	0.99	0.99	0.99	0.98	0.98	0.98
20°	0.96	0.97	0.98	0.98	0.97	0.97	0.97	0.96
25°	0.93	0.95	0.96	0.96	0.95	0.95	0.95	0.94
30°	0.91	0.92	0.94	0.94	0.93	0.93	0.93	0.91
35°	0.88	0.90	0.92	0.91	0.90	0.90	0.90	0.89
40°	0.84	0.87	0.89	0.88	0.87	0.87	0.87	0.85
45°	0.80	0.83	0.86	0.85	0.83	0.84	0.84	0.82
50°	0.76	0.80	0.83	0.82	0.79	0.80	0.81	0.78
55°	0.72	0.76	0.80	0.78	0.75	0.76	0.78	0.75
60°	0.67	0.72	0.76	0.74	0.70	0.73	0.74	0.71
65°	0.63	0.68	0.73	0.70	0.66	0.69	0.71	0.67

More Than One Type of Material and/or Skylight

$$RTTV = \frac{12.5\,(A_{r1}xU_{r1} + A_{r2}xU_{r2} + ..+ A_{rn}xU_{rn})}{A_o}$$

$$+\ 4.8\frac{(A_{s1}xU_{s1} + A_{s2}xU_{s2} + ...+ A_{sn}xU_{sn})}{A_o} +$$

$$485\frac{(A_{s1}xSC_{s1} + A_{s2}xSC_{s2} + ...+ A_{sn}xSC_{sn})}{A_o}(CF)$$

A_{r1}, A_{r2}, A_{rn} = areas of different opaque roof (m^2)

A_{s1}, A_{s2}, A_{sn} =areas of different skylight (m^2)

A_o =gross area of roof (m^2)

U_{r1}, U_{r2}, U_{rn}= thermal transmittances of opaque roofs (W/m^2K)

U_{s1}, U_{s2}, U_{sn}= shading coefficients of skylights

SC_{s1}, SC_{s2}, SC_{sn} = thermal transmittances of skylights (W/m^2K)

Sections Facing Different Orientations or Pitched at Different Angles

$$RTTV = \frac{A_{o1} \times RTTV_1 + A_{o2} \times RTTV_2 + \ldots + A_{on} \times RTTV_n)}{A_{o1} + A_{o2} + \ldots\ldots + A_{on}}$$

Where :

A_{o1}, A_{o2} , A_{on} are the gross areas of the roof for each section (m^2)

RETV

$$ETTV = 12\ (1\text{-}WWR)U_w + 3.4(WWR)\ U_f + 211(WWR)(CF)(SC)$$

Similar to concept of ETTV and RTTV except that the formula is modified to:

$$RETV = 3.4(1\text{-}WWR)U_w + 1.3(WWR)\ U_f + 58.6(WWR)(CF)(SC)$$

Maximum permissible **RETV** has been set at **25 W/m²**.

Solar Correction Factor – Walls (RETV)

Table C3 : Solar Correction Factors (CF) for Walls (RETV)

Pitch Angle	Orientation							
	N	NE	E	SE	S	SW	W	NW
90°	0.83	1.01	1.18	1.02	0.86	1.09	1.26	1.06
80°	0.98	1.15	1.34	1.18	1.01	1.23	1.41	1.21
70°	1.13	1.30	1.48	1.35	1.17	1.38	1.55	1.36

Note: 1. The correction factors for other orientations and other pitch angles may be obtained by interpolation.

RETV

Deem-to-Satisfy Criteria for RETV

Building's WWR and SC fall within any one of following sets of criteria, building deemed to have complied with RETV and; hence, exempted from computing RETV.

WWR Bldg < 0.3 and SC facade < 0.7 or
WWR Bldg < 0.4 and SC facade < 0.5 or
WWR Bldg < 0.5 and SC facade < 0.43
where
$SC = SC_{Glass} x SC_{shading\ device}$.
SC of each facade must not exceed the prescribed value.
Note: Applicable to buildings with external masonry walls.

Example

Figures E1 and E2 show the typical floor plan and one elevation view of a building. All external walls and windows are at 90° to the horizontal floor.

The external opaque walls enclosing air-conditioned spaces of the building are constructed of 15 mm granite panels on the outside, 100 mm light weight concrete (1120kg/m^3), 20 mm air space and 20 mm gypsum plaster board on the inside. The inside surface of the opaque walls can be assumed to have high emissivity.

Example

The windows are double-glazed with a shading coefficient of 0.3 and U-value of 2.8 W/m^{2}K. There is no external shading for all the windows.

ETTV = 12(1 - WWR)Uw + 3.4(WWR)Uf + 211 (WWR)(CF)(SC)

Using the information given in Figures E3, E4, E5 and E6, determine:

(i) the overall U-value of the opaque wall,
(ii) ETTV of Façade A,
(iii) ETTV of the whole building and state whether it meets the requirement

Example

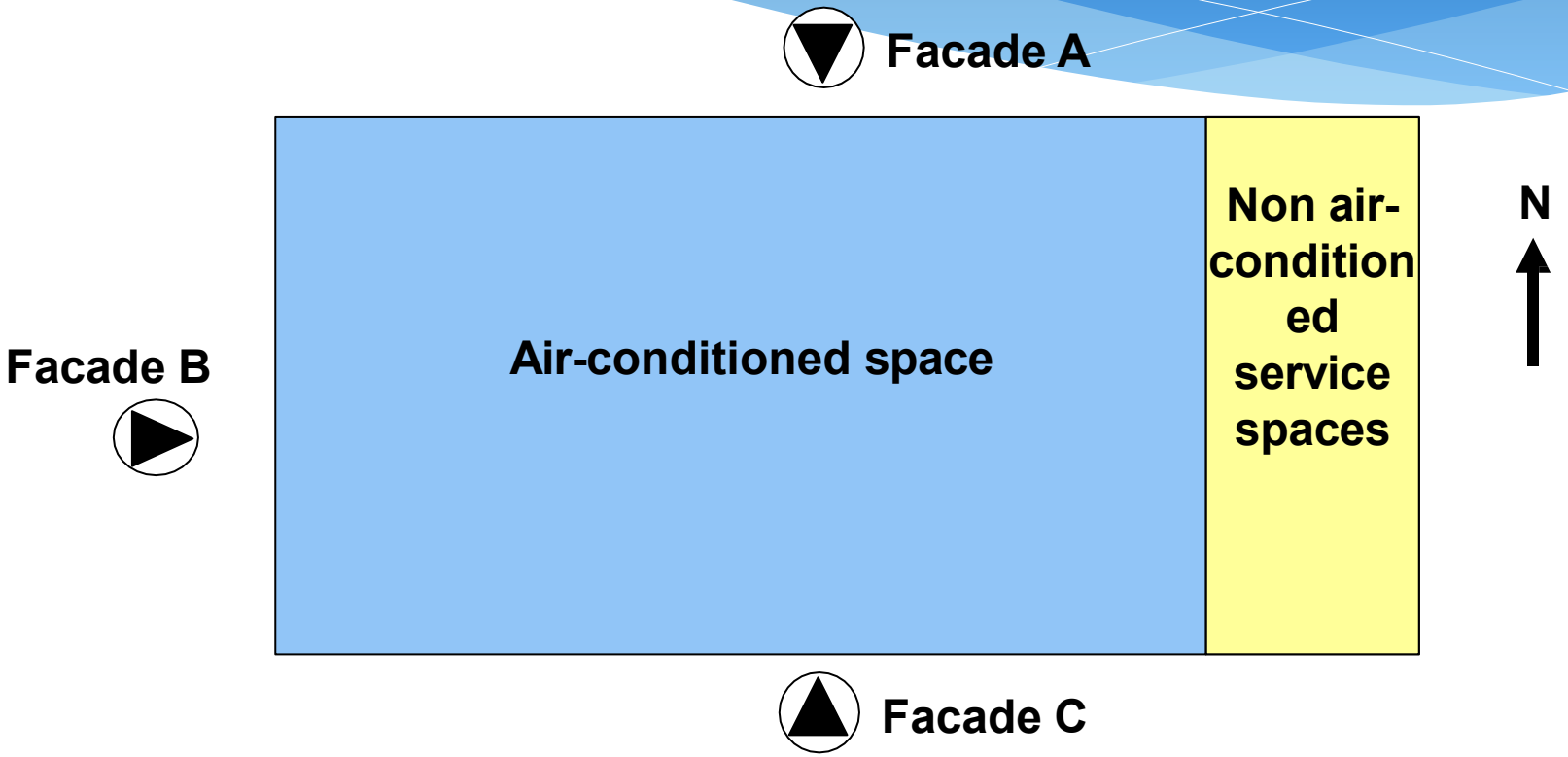


Figure E1 – Typical Floor Plan

Example

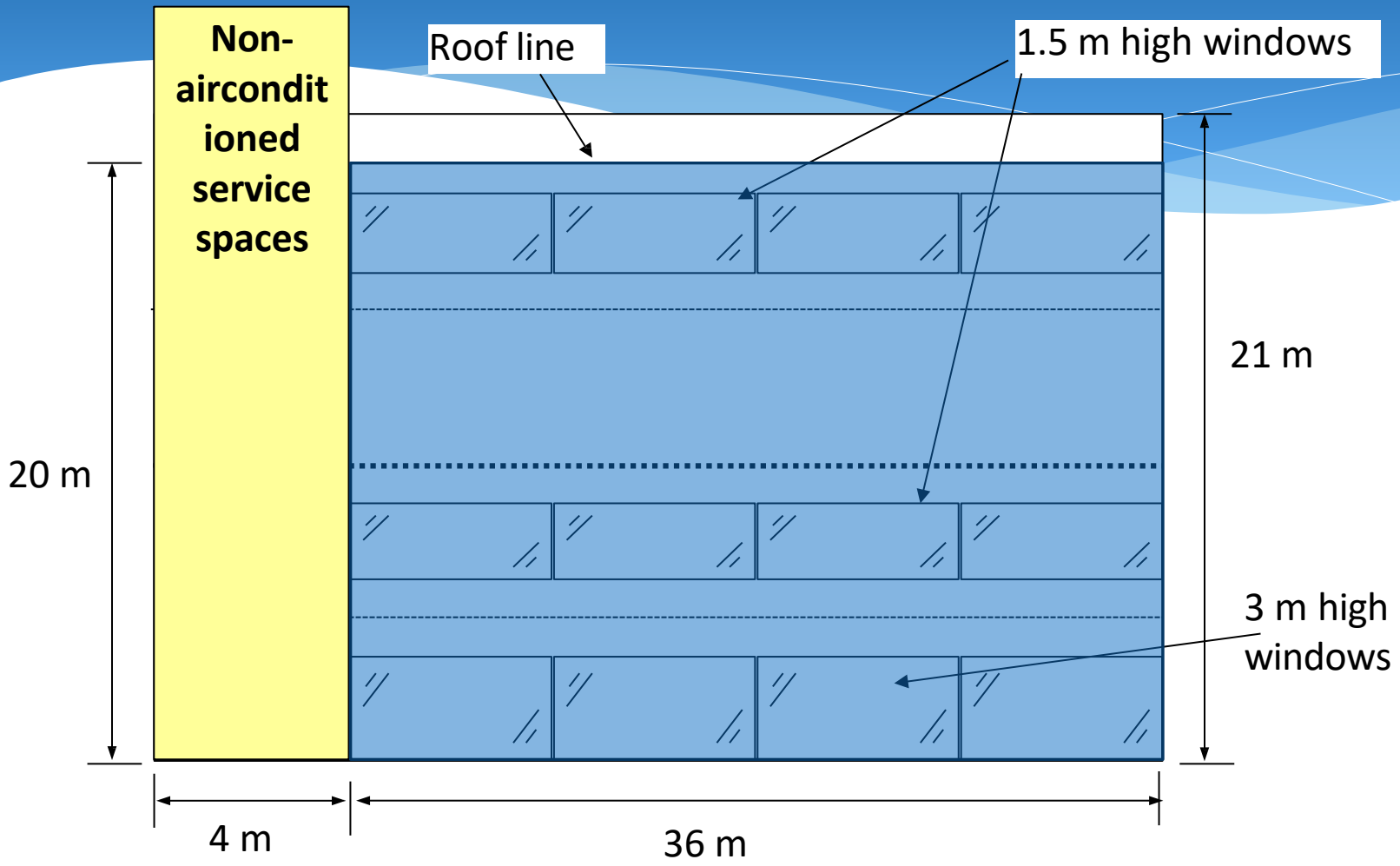


Figure E2 – Façade A

Example

Material	Density (kg/m³)	K-value (W/mK)
Concrete, light weight	1120	0.346
Gypsum plaster board	880	0.17
Glass, sheet	2512	1.053
Granite	2640	2.927

Figure E3 - Thermal Conductivity Values of Some Basic Materials

Type of Surface/Air Space	Thermal Resistance (m² K/W)
Surface Film Resistances for Walls	
1. Inside surface (R_i)	0.120
(a) High emissivity	0.299
(b) Low emissivity	
2. Outside surface (R_o)	0.044
- High emissivity	
20 mm air space	0.148

Figure E4 - Surface Film and Air Space Resistances for Walls

Example

Pitch Angle	Orientation							
	N	NE	E	SE	S	SW	W	NW
85°	0.89	1.05	1.21	1.07	0.92	1.14	1.31	1.11
90°	0.80	0.97	1.13	0.98	0.83	1.06	1.23	1.03
95°	0.73	0.90	1.05	0.91	0.76	0.99	1.15	0.96

Figure E5 – Solar Correction Factors for Walls

Facade	Gross External Wall Area (m^2)	ETTV (W/m^2)
B	400	60
C	720	35

Figure E6 – Gross External Wall Areas and ETTVs

Official Open

Example - Solution

(i) U-value of Opaque Wall

$$R_T = R_o + \frac{x_1}{k_1} + \frac{x_2}{k_2} + R_a + \frac{x_3}{k_3} + R_i$$

From Fig E4, $R_o = 0.044\ m^2K/W$ & $R_i = 0.12\ m^2K/W$

$R_a = 0.148\ m^2K/W$

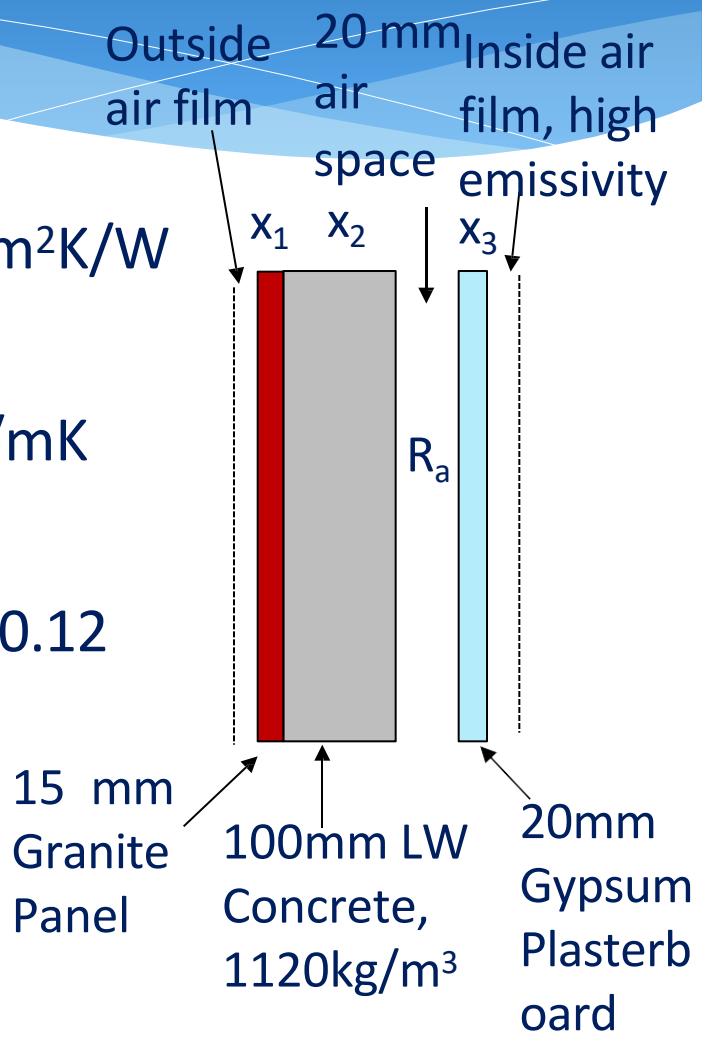
From Fig E3, $k_1 = 2.927 W/mK$, $k_2 = 0.346 W/mK$

$k_3 = 0.17 W/mK$

$$R_T = 0.044 + \frac{0.015}{2.927} + \frac{0.10}{0.346} + 0.148 + \frac{0.02}{0.17} + 0.12$$

$$= 0.724\ m^2K/W$$

$$U_T = \frac{1}{R_T} = \frac{1}{0.724} = 1.38\ W/m^2K$$



Example - Solution

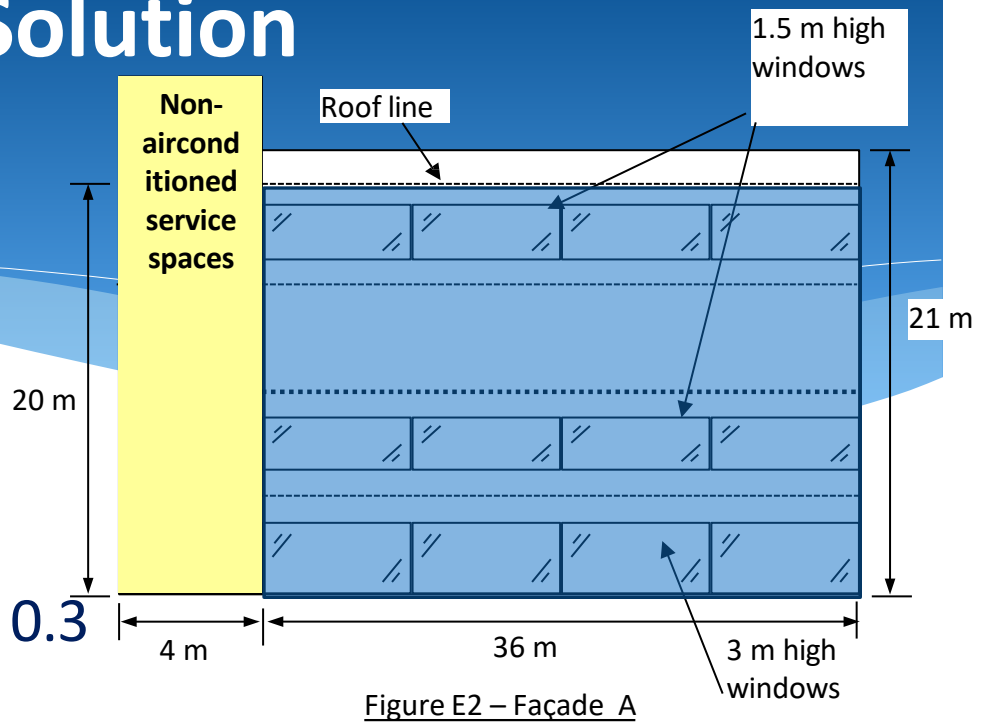


Figure E2 – Façade A

(ii) ETTV of Façade A

$$WWR = \frac{(36x1.5x2 + 36x3)}{36 \times 20} = \frac{216}{720} = 0.3$$

From Figure E5, North, 90°, CF = 0.8

$$ETTV = 12(1 - WWR)U_w + 3.4(WWR)U_f + 211\,(WWR)(CF)(SC)$$

$$= 12(1 - 0.3)\,1.38 + 3.4\,(0.3)\,2.8 + 211\,(0.3)\,(0.8)\,(0.3)$$

$$= 11.592 + 2.856 + 15.192$$

$$= 29.64 W/m^2$$

Example - Solution

(iii) ETTV of Whole Building

$$\text{Overall ETTV} = \frac{ETTV_N.A_N + ETTV_S.A_S + ETTV_W.A_W}{A_N + A_S + A_W}$$

$$= \frac{29.64x720 + 60x400 + 35x720}{720 + 400 + 720}$$

$$= 38.33\ W/m^2$$

Since the ETTV of 38.33 W/m^2 is less than the maximum permissible value of 50 W/m^2, it complies with the requirements.

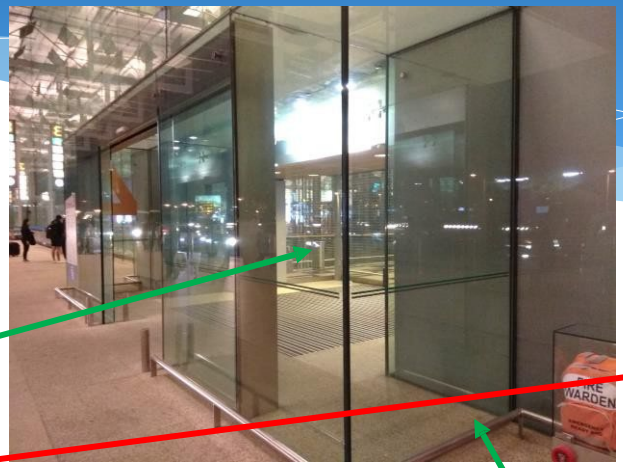
Air Infiltration

* Reduce Outdoor Air Infiltration
  * Use of weatherstripping materials around doors and windows
  * Seal up cracks in building wall



Air Infiltration

* Reduce Outdoor Air Infiltration
  * Automatic closing doors/double doors/rotating doors
  * Air curtains









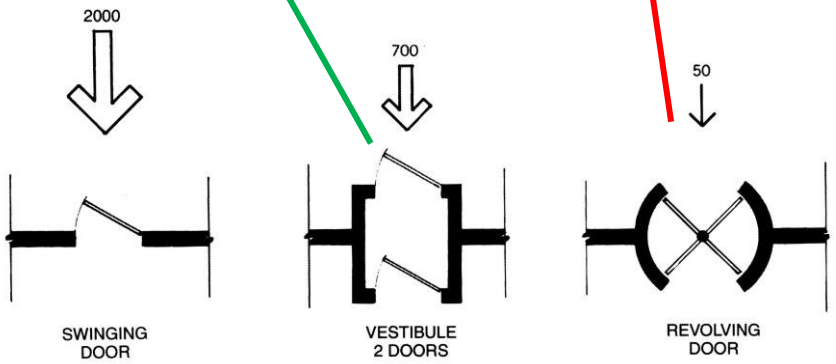


Figure 15.13b The number indicates the cubic feet of air infiltrating due to one-door operation.